Scenario Brief - Activity 1

Situation

A new process engineer must explain why doping, junction bias and oxide thickness matter before reviewing a CMOS traveller.

Your role

You are the process-integration engineer preparing evidence for a cross-functional review. Device, process, yield, reliability and manufacturing stakeholders will challenge unsupported claims.

Operating constraints

- Protect product and customer confidentiality; use only the supplied or de-identified information.
- Separate facts from assumptions.
- Do not release or restart a process without objective criteria.
- Prefer the smallest safe test that can distinguish competing explanations.

Review questions

1. What is directly observed?
2. Which process modules or device structures could plausibly create the observation?
3. What evidence supports or contradicts each mechanism?
4. What is the immediate risk to wafers, devices, yield, performance or reliability?
5. What decision is justified now, and what evidence is still needed?